

OLGA CONNECT

Database Architecture and Table-Level Design

Overall product, mobile application, NLP matching, events, consent, connections, chat, moderation and operations

Document	Value
Status	Final draft for architecture review and engineering implementation
Version	2.2 - event and notification policy completion
Target	QA-ready Azure test environment by 15 October 2026
Database	Azure SQL Database; schema-separated modular monolith for Release 1
Source baseline	Product & Technical Blueprint; NLP Service Architecture; NLP Matching Technical Summary; current .NET/Azure SQL implementation
Scope	58 proposed tables plus 4 controlled views

Architecture correction The product database is not NLP-centric. NLP is one bounded domain. Member identity, profile, consent, event presence, relationships and moderation remain authoritative product domains; NLP consumes approved projections and cannot grant visibility or communication rights.

1. Executive architecture decision

Implement OLGA Connect Release 1 on one Azure SQL Database per environment, separated into domain schemas and accessed only through service APIs. This aligns with the existing .NET NLP implementation and keeps the October 15 delivery practical while preserving clear ownership boundaries for a later service split.

Area	Decision	Release 1 rationale
Transactional store	Azure SQL Database	Matches existing SQL scripts/EF Core code; strong transactions for consent, connections and chat authorization.
Large objects	Azure Blob Storage	Private files, exports and evaluation reports remain outside OLTP rows.
Mobile local state	Encrypted SQLite/cache	Offline queue and recent read models only; Azure SQL remains authoritative.
Async and realtime	Service Bus + SignalR or equivalent	Embedding, scans and notifications are decoupled; chat/request updates are realtime.
Identity/secrets	Microsoft Entra/approved CIAM + managed identity + Key Vault	No credentials or model keys in mobile code.
NLP integration	Read-only eligibility views + owned nlp schema	Existing matching service keeps its boundary without duplicating product truth.

2. Source requirements incorporated

Source	Key requirements carried into this design
OLGA Connect Product, NLP and Technical Blueprint v1.0	Registration/verification; profiles; organizations/sectors/geography; semantic recommendations; event check-in/Live Mode; connection consent gate; chat/files; moderation; analytics; privacy rights.
OLGA Connect NLP Service Architecture	Standalone API; bounded candidate repository; original/normalized text; embeddings; reciprocal scoring; deterministic reasons; feedback/evaluation; EF Core + stored procedures.
OLGA NLP Matching - Technical Summary	Precompute embeddings; Azure SQL eligibility and vector similarity; idempotent requests; model/preprocessing/ranking versions; 3-7 explained matches.
Current .NET/Azure SQL repository	Existing nlp.NlpIntent, NlpEmbedding, NlpRankingConfig, NlpModelVersion, NlpMatchResult, NlpFeedback and temporary eligibility/relationship tables.

3. Design principles

Principle	Application
Consent before capability	A recommendation never grants access; connection acceptance unlocks chat and file exchange.

Principle	Application
One owner per fact	Member, consent, event presence, relationship, chat and NLP data each have an authoritative owner.
No direct client database access	Mobile/admin applications call authenticated APIs; service identities receive least privilege.
Privacy-minimized context	Use coarse, short-lived presence and business geography; never expose exact member location.
Explainable/versioned NLP	Persist model, preprocessing, ranking, scores and approved reason codes for every result.
Transactional safety	Idempotency, rowversion, foreign keys and outbox events protect mobile retries and integration.
Modular monolith first	One database with schema boundaries reduces delivery risk; split only when scale/team ownership requires it.
Measure before adding infrastructure	Redis, dedicated vector search and separate analytics stores require demonstrated need.

4. Overall data architecture

OLGA Connect - Overall Data Architecture

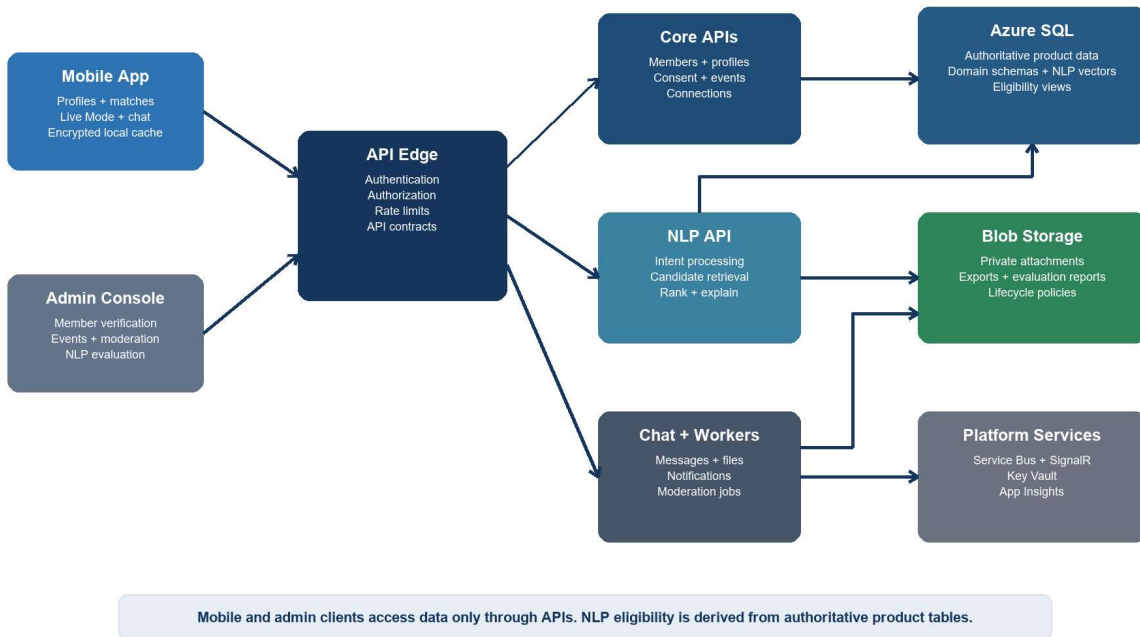


Figure 1. Overall product and data architecture

OLGA Connect - Domain Data Model

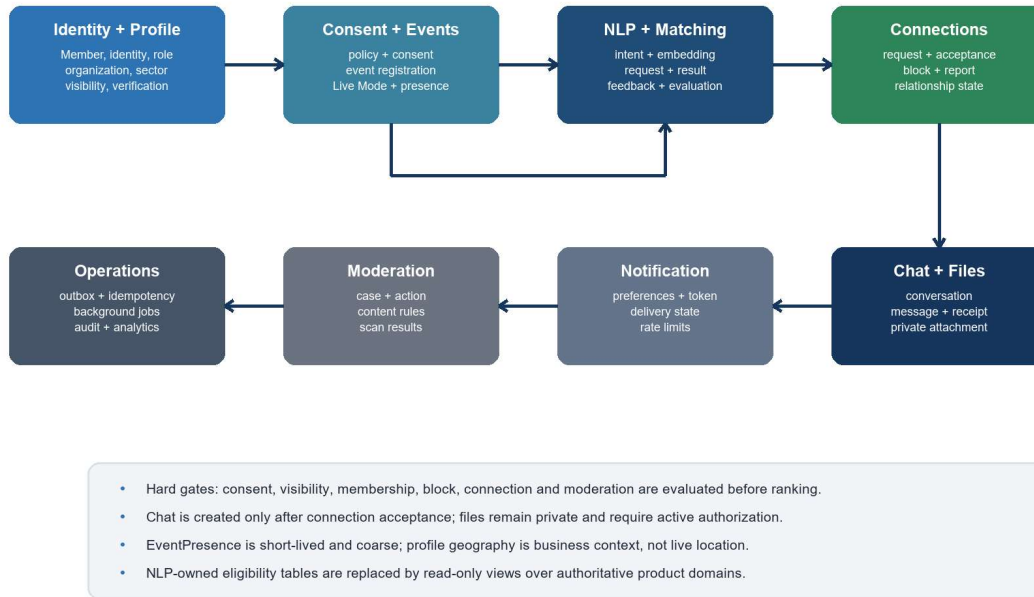


Figure 2. Domain ownership and relationship model

5. Schema ownership and access

Schema	Tables	Primary owner	Access rule
core	Community, Organization, OrganizationMember, MemberProfile, Sector, MemberSector, MemberGeography, ProfileFieldVisibility, MemberVerification	Core API	Read/write only by owning service; cross-domain reads through API or approved view.
iam	Member, MemberIdentity, Role, MemberRole, MemberDevice	Identity/Core API	Read/write only by owning service; cross-domain reads through API or approved view.
consent	ConsentPolicy, MemberConsent, PrivacyRequest	Privacy/Product admin	Read/write only by owning service; cross-domain reads through API or approved view.
event	Venue, Event, EventMatchingPolicy, EventRegistration, LiveModeSession, EventPresence	Event admin	Read/write only by owning service; cross-domain reads through API or approved view.
social	ConnectionRequest, Connection, MemberBlock, MemberReport	Social API	Read/write only by owning service; cross-domain reads through API or approved view.
chat	Conversation, ConversationParticipant, Message, Attachment, MessageReceipt	Chat service	Read/write only by owning service; cross-domain reads through API or approved view.

Schema	Tables	Primary owner	Access rule
notification	NotificationPolicy, NotificationPreference, PushToken, Notification, NotificationDeliveryAttempt	Notification admin	Read/write only by owning service; cross-domain reads through API or approved view.
nlp	NlpIntent, NlpEmbedding, NlpModelVersion, NlpRankingConfig, NlpProcessingJob, MatchRequest, NlpMatchResult, NlpFeedback, MatchSuppression, EvaluationDataset, EvaluationPair, EvaluationRun	NLP service	Read/write only by owning service; cross-domain reads through API or approved view.
moderation	ModerationCase, ModerationAction, ContentRule, ContentScan	Moderation service	Read/write only by owning service; cross-domain reads through API or approved view.
ops	OutboxEvent, IdempotencyRecord, BackgroundJob, AuditEvent	Owning service	Read/write only by owning service; cross-domain reads through API or approved view.
analytics	ProductEvent	Analytics pipeline	Read/write only by owning service; cross-domain reads through API or approved view.

NLP boundary Remove production ownership of nlp.MemberContextEligibility and nlp.MemberRelationship. Replace them with read-only views derived from iam/core/consent/event/social tables. Local test fixtures may retain stand-alone tables behind the repository interface.

6. Critical relationship and transaction rules

Workflow	Invariant
Member activation	Requires verified identity, active profile, accepted mandatory policy and no active suspension.
Match eligibility	Active member/profile + visibility + valid intent/embedding + context eligibility + consent + no block + no existing connection + no suppression.
Live Mode	Requires active event, registration/check-in as configured, current consent and non-expired LiveModeSession.
Event match alert	Requires active EventMatchingPolicy, event eligibility, high-confidence match, fresh permitted proximity when configured, member preference and the stricter event/channel frequency cap.
Notification enqueue	Resolves immutable NotificationPolicy, applies opt-out, quiet hours, dedupe, expiry and rate limits atomically, then publishes through the transactional outbox.
Connection acceptance	Atomically updates request, creates canonical Connection, creates Conversation/participants and writes OutboxEvent.
Chat send	Requires active connection, participant authorization, no block/restriction and idempotent message ID.
File availability	Requires clean upload, clean/type validation and CLEAN clean scan before message attachment becomes downloadable.
Account deletion	Disables identity/session immediately, removes discovery, closes Live Mode and schedules anonymization/purge per policy.

7. Controlled views and procedures

Object	Source	Output	Consumer
nlp.vw_MemberContextEligibility	Member + profile + consent + event registration/Live Mode projection	member_id, context_id, is_live, is_visible, has_consent, is_suspended, is_deleted	NLP Candidate Repository read-only
nlp.vw_MemberRelationship	Block and accepted connection projection	member_id, other_member_id, context_id, is_blocked, is_connected	NLP Candidate Repository read-only
chat.vw_AuthorizedConversation	Active connection and participant authorization projection	conversation_id, member_id, other_member_id, status	Chat authorization checks
admin.vw_MemberReview	Redacted member/profile/verification/moderation projection	member_id, profile_status, verification_status, open_case_count	Admin console

Stored procedures should be narrow and version-controlled. Keep `nlp.GetRequesterIntent` and `nlp.GetEligibleCandidates` for bounded eligibility-aware retrieval; `nlp.SaveMatchResults` for atomic idempotent persistence; `social.AcceptConnectionRequest` for connection/chat creation; `chat.SaveMessage` for authorization plus idempotency; `event.PurgeExpiredPresence` for retention enforcement; and `notification.TryEnqueue` for atomic policy, preference, threshold, dedupe, quiet-hours and frequency-cap enforcement.

8. Table-level data dictionary

This section is the implementation baseline. SQL types are Azure SQL types. FK delete behavior defaults to NO ACTION unless explicitly stated; business deletion is performed through controlled workflows. Every service query must include its authorization/community/context boundary.

9. CORE schema

core.Community

Purpose: Network/community boundary; one OLGA row in Release 1. **Owner:** Core API **Classification:** Internal **Retention:** Life of community + audit period

Column	SQL type	Null	Key	Default	Description
community_id	varchar(64)	No	PK		Stable API identifier.
name	nvarchar(200)	No			Display name.
status	varchar(24)	No		'ACTIVE'	ACTIVE, SUSPENDED or CLOSED.
default_locale	varchar(16)	No		'en'	BCP-47 default locale.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: UQ(name); IX(status)

Rules: Do not implement full tenant isolation for Release 1; retain community_id so the boundary is explicit.

core.Organization

Purpose: Professional organization represented in profiles. **Owner:** Core API **Classification:** Business public/confidential **Retention:** While referenced + archive period

Column	SQL type	Null	Key	Default	Description
organization_id	varchar(64)	No	PK		Stable identifier.
community_id	varchar(64)	No	FK		FK core.Community.
name	nvarchar(250)	No			Canonical organization name.
normalized_name	nvarchar(250)	No			Search/deduplication form.

Column	SQL type	Null	Key	Default	Description
website_domain	nvarchar(255)	Yes			Verified domain when available.
industry_code	varchar(64)	Yes	FK		Reference taxonomy.
verification_status	varchar(24)	No		'UNVERIFIED'	UNVERIFIED, PENDING, VERIFIED, REJECTED.
status	varchar(20)	No		'ACTIVE'	Lifecycle state.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: IX(community_id,normalized_name); IX(website_domain)

Rules: Potential duplicates require moderation merge, not silent deletion.

core.OrganizationMember

Purpose: Member affiliation, title and organization-level role. **Owner:** Core API **Classification:** Personal/business **Retention:** Profile life + approved history

Column	SQL type	Null	Key	Default	Description
organization_member_id	bigint	No	PK	IDENTITY	Internal key.
organization_id	varchar(64)	No	FK		FK core.Organization.
member_id	varchar(64)	No	FK		FK iam.Member.
job_title	nvarchar(200)	Yes			Professional title.
department	nvarchar(150)	Yes			Optional function/department.
is_primary	bit	No		0	Primary current affiliation.
started_on	date	Yes			Optional month/day precision per product policy.
ended_on	date	Yes			Null for current affiliation.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: UQ(organization_id,member_id,started_on); IX(member_id,is_primary)
Rules: Profile visibility rules control exposure.

core.MemberProfile

Purpose: Searchable professional profile and visibility state. **Owner:** Core API **Classification:** Personal/business **Retention:** Account life + deletion policy

Column	SQL type	Null	Key	Default	Description
member_id	varchar(64)	No	PK/FK		One-to-one FK iam.Member.
display_name	nvarchar(150)	No			Member-facing name.
headline	nvarchar(240)	Yes			Short professional headline.
professional_summary	nvarchar(2000)	Yes			Member-authored summary.
role_category	varchar(64)	Yes			Controlled role/function code.
profile_status	varchar(24)	No		'DRAFT'	DRAFT, PENDING_REVIEW, ACTIVE, HIDDEN.
visibility	varchar(20)	No		'MEMBERS'	PRIVATE, MEMBERS or CONTEXT_ONLY.
completeness_score	decimal(5,2)	No		0	Derived profile completeness percentage.
published_at	datetimeoffset(7)	Yes			First/current publication time.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: IX(profile_status,visibility); IX(updated_at)
Rules: Contact details are never stored in public profile fields.

core.Sector

Purpose: Controlled sector/industry taxonomy. **Owner:** Product admin **Classification:** Public reference **Retention:** Indefinite/versioned

Column	SQL type	Null	Key	Default	Description
sector_code	varchar(64)	No	PK		Stable taxonomy code.
parent_sector_code	varchar(64)	Yes	FK		Self-referencing hierarchy.
name	nvarchar(150)	No	UQ		Display label.

Column	SQL type	Null	Key	Default	Description
status	varchar(16)	No		'ACTIVE'	ACTIVE or RETIRED.
sort_order	int	No		0	Presentation order.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: UQ(name); IX(parent_sector_code,status)

Rules: Retire codes rather than deleting referenced taxonomy.

core.MemberSector

Purpose: Many-to-many profile sector selection. **Owner:** Core API **Classification:** Business **Retention:** Profile life

Column	SQL type	Null	Key	Default	Description
member_id	varchar(64)	No	PK/FK		FK MemberProfile.
sector_code	varchar(64)	No	PK/FK		FK Sector.
is_primary	bit	No		0	Primary sector flag.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Assignment time.

Indexes: PK(member_id,sector_code); IX(sector_code,member_id)

Rules: At most one primary sector per member, enforced by filtered unique index.

core.MemberGeography

Purpose: Member operating markets; not live location. **Owner:** Core API **Classification:** Business **Retention:** Profile life

Column	SQL type	Null	Key	Default	Description
member_geography_id	bigint	No	PK	IDENTITY	Internal key.
member_id	varchar(64)	No	FK		FK MemberProfile.
country_code	char(2)	No			ISO 3166-1 alpha-2.
region	nvarchar(120)	Yes			State/region.

Column	SQL type	Null	Key	Default	Description
city	nvarchar(120)	Yes			Operating city.
is_primary	bit	No		0	Primary market.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Creation time.

Indexes: IX(member_id,is_primary); IX(country_code,region,city)

Rules: Do not reuse for proximity; it is profile/business geography.

core.ProfileFieldVisibility

Purpose: Per-field exposure before/after connection. **Owner:** Core API **Classification:** Confidential **Retention:** Profile life + consent audit

Column	SQL type	Null	Key	Default	Description
member_id	varchar(64)	No	PK/FK		FK MemberProfile.
field_code	varchar(64)	No	PK		PROFILE_SUMMARY, ORGANIZATION, GEOGRAPHY, etc.
audience	varchar(24)	No		'CONNECTED'	PRIVATE, MATCHED, CONNECTED, MEMBERS.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Last preference change.
row_version	rowversion	No	Concurrency	Generated	Concurrency token.

Indexes: PK(member_id,field_code); IX(audience)

Rules: Server applies this policy to every profile response.

core.MemberVerification

Purpose: Administrator/member verification workflow. **Owner:** Admin/Core API **Classification:** Restricted **Retention:** Account life + audit period

Column	SQL type	Null	Key	Default	Description
verification_id	varchar(64)	No	PK		Case identifier.
member_id	varchar(64)	No	FK		FK iam.Member.
verification_type	varchar(32)	No			EMAIL, PHONE, ORGANIZATION or MANUAL.
status	varchar(24)	No		'PENDING'	PENDING, APPROVED, REJECTED, EXPIRED.
evidence_attachment_id	varchar(64)	Yes	FK		Private attachment reference, if policy permits.

Column	SQL type	Null	Key	Default	Description
reviewed_by	varchar(64)	Yes	FK		Administrator member ID.
reviewed_at	datetimeoffset(7)	Yes			Review completion.
reason_code	varchar(64)	Yes			Controlled result reason.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: IX(member_id,status); IX(status,created_at)

Rules: Evidence must not be exposed through member APIs.

10. IAM schema

iam.Member

Purpose: Authoritative member account and lifecycle state. **Owner:** Identity/Core API **Classification:** Personal **Retention:** Account life + approved legal/audit period

Column	SQL type	Null	Key	Default	Description
member_id	varchar(64)	No	PK		Opaque identifier; use GUID/ULID string.
community_id	varchar(64)	No	FK		FK core.Community.
status	varchar(24)	No		'PENDING'	PENDING, ACTIVE, SUSPENDED, ANONYMIZED, DELETED.
locale	varchar(16)	No		'en'	Preferred locale.
verified_at	datetimeoffset(7)	Yes			First completed account verification.
suspended_at	datetimeoffset(7)	Yes			Administrative suspension time.
deleted_at	datetimeoffset(7)	Yes			Deletion/anonymization workflow start.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: IX(community_id,status); IX(status,updated_at)

Rules: External identity claims map through MemberIdentity; never expose predictable identifiers.

iam.MemberIdentity

Purpose: Email/mobile/passwordless/Entra identity mapping. **Owner:** Identity service **Classification:** Restricted personal **Retention:** Account life; hash/audit per policy

Column	SQL type	Null	Key	Default	Description
member_identity_id	bigint	No	PK	IDENTITY	Internal key.
member_id	varchar(64)	No	FK		FK iam.Member.
provider	varchar(32)	No			EMAIL, PHONE, ENTRA or approved provider.
provider_subject	nvarchar(320)	No	UQ		Normalized email/mobile or external subject; encrypt/tokenize if required.
is_primary	bit	No		0	Primary sign-in identity flag.
verified_at	datetimeoffset(7)	Yes			Verification completion.
last_login_at	datetimeoffset(7)	Yes			Security/account support signal.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Creation time.

Indexes: UQ(provider,provider_subject); IX(member_id,is_primary)

Rules: Do not store passwords or OTP values; keep short-lived verification secrets in the identity platform/cache.

iam.Role

Purpose: Controlled role catalog for member and administrator authorization. **Owner:** Identity service **Classification:** Internal **Retention:** Indefinite while referenced

Column	SQL type	Null	Key	Default	Description
role_code	varchar(64)	No	PK		MEMBER, ADMIN, MODERATOR, NLP_EVALUATOR.
name	nvarchar(100)	No	UQ		Display name.
description	nvarchar(500)	Yes			Scope and intended use.
is_privileged	bit	No		0	Requires elevated authentication and audit.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: UQ(name)**Rules:** Seed through migrations; role changes are audited.

iam.MemberRole

Purpose: Role assignment with optional expiry. **Owner:** Identity service **Classification:** Confidential **Retention:** Assignment life + audit period

Column	SQL type	Null	Key	Default	Description
member_id	varchar(64)	No	PK/FK		FK iam.Member.
role_code	varchar(64)	No	PK/FK		FK iam.Role.
granted_by	varchar(64)	Yes	FK		Administrator member ID.
granted_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Grant time.
expires_at	datetimeoffset(7)	Yes			Optional expiry.
revoked_at	datetimeoffset(7)	Yes			Revocation time.

Indexes: PK(member_id,role_code); IX(role_code,expires_at)**Rules:** Privileged assignment/revocation writes AuditEvent.

iam.MemberDevice

Purpose: Registered mobile installation and security state. **Owner:** Identity/Mobile API **Classification:** Confidential **Retention:** Until logout/revocation + 30 days

Column	SQL type	Null	Key	Default	Description
device_id	varchar(64)	No	PK		Client installation identifier, rotated on reinstall.
member_id	varchar(64)	No	FK		FK iam.Member.
platform	varchar(16)	No			IOS or ANDROID.
app_version	varchar(32)	No			Last reported app version.
status	varchar(20)	No		'ACTIVE'	ACTIVE or REVOKED.
last_seen_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Last authenticated request.
revoked_at	datetimeoffset(7)	Yes			Remote logout/device loss handling.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.

Column	SQL type	Null	Key	Default	Description
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: IX(member_id,status); IX(last_seen_at)

Rules: Authentication tokens stay in the OS secure keystore, never this table.

11. CONSENT schema

consent.ConsentPolicy

Purpose: Versioned consent purpose and legal/community text. **Owner:** Privacy/Product admin **Classification:** Internal/public text **Retention:** Indefinite/versioned

Column	SQL type	Null	Key	Default	Description
policy_id	varchar(64)	No	PK		Stable policy version ID.
purpose_code	varchar(64)	No			TERMS, LOCATION, LIVE_MODE, NOTIFICATIONS, ANALYTICS.
version	varchar(32)	No			Human-readable version.
locale	varchar(16)	No		'en'	Text locale.
content_hash	char(64)	No			SHA-256 of presented content.
effective_from	datetimeoffset(7)	No			Activation time.
retired_at	datetimeoffset(7)	Yes			Retirement time.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: UQ(purpose_code,version,locale); IX(purpose_code,effective_from)

Rules: Published policy text is immutable.

consent.MemberConsent

Purpose: Authoritative grant/withdrawal evidence by purpose. **Owner:** Privacy/Core API **Classification:** Restricted personal **Retention:** Per legal/privacy policy

Column	SQL type	Null	Key	Default	Description
member_consent_id	bigint	No	PK	IDENTITY	Evidence record.

Column	SQL type	Null	Key	Default	Description
member_id	varchar(64)	No	FK		FK iam.Member.
policy_id	varchar(64)	No	FK		FK ConsentPolicy.
decision	varchar(16)	No			GRANTED or DENIED.
captured_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Decision time.
withdrawn_at	datetimeoffset(7)	Yes			Withdrawal time.
capture_channel	varchar(24)	No			MOBILE, WEB_ADMIN or SUPPORT.
evidence_json	nvarchar(1000)	Yes			Bounded device/app/version evidence; no secrets.

Indexes: IX(member_id,policy_id,captured_at DESC); IX(policy_id,decision)

Rules: Append decisions; do not overwrite historical evidence. Current consent is derived from latest valid record.

consent.PrivacyRequest

Purpose: Member data access, correction, deletion or consent-support case. **Owner:** Privacy/Admin **Classification:** Highly restricted **Retention:** Per regulatory policy

Column	SQL type	Null	Key	Default	Description
privacy_request_id	varchar(64)	No	PK		Case identifier.
member_id	varchar(64)	No	FK		Requesting member.
request_type	varchar(24)	No			ACCESS, CORRECT, DELETE, EXPORT, CONSENT_SUPPORT.
status	varchar(24)	No		'OPEN'	OPEN, VERIFIED, PROCESSING, COMPLETED, REJECTED.
verified_at	datetimeoffset(7)	Yes			Identity verification.
due_at	datetimeoffset(7)	Yes			Policy deadline.
completed_at	datetimeoffset(7)	Yes			Completion time.
result_attachment_id	varchar(64)	Yes	FK		Private export package reference.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: IX(status,due_at); IX(member_id,created_at DESC)

Rules: Every state transition and export access is audited.

12. EVENT schema

event.Venue

Purpose: Admin-configured venue and coarse proximity boundary. **Owner:** Event admin **Classification:** Business/location **Retention:** While event is active + archive

Column	SQL type	Null	Key	Default	Description
venue_id	varchar(64)	No	PK		Venue identifier.
name	nvarchar(200)	No			Venue name.
country_code	char(2)	No			ISO country.
region	nvarchar(120)	Yes			Region/state.
city	nvarchar(120)	Yes			City.
coarse_geo_cell	varchar(32)	Yes			Venue-level geohash/H3 cell; not member location.
timezone_id	varchar(64)	No			IANA/Windows mapping controlled by service.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: IX(country_code,region,city)

Rules: Exact coordinates, if required for geofencing, are restricted configuration and never returned to members.

event.Event

Purpose: Event/context used for registration, Live Mode and matching. **Owner:** Event admin **Classification:** Business **Retention:** Event life + archive period

Column	SQL type	Null	Key	Default	Description
event_id	varchar(64)	No	PK		Also used as matching context_id.
community_id	varchar(64)	No	FK		FK Community.
venue_id	varchar(64)	Yes	FK		FK Venue.

Column	SQL type	Null	Key	Default	Description
name	nvarchar(250)	No			Event name.
description	nvarchar(2000)	Yes			Admin-managed description.
starts_at	datetimeoffset(7)	No			UTC start.
ends_at	datetimeoffset(7)	No			UTC end.
status	varchar(24)	No		'DRAFT'	DRAFT, PUBLISHED, ACTIVE, COMPLETED, CANCELLED.
live_mode_enabled	bit	No		0	Event allows Live Mode.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: IX(community_id,status,starts_at); IX(venue_id,starts_at)

Rules: CHECK ends_at > starts_at.

event.EventMatchingPolicy

Purpose: Versioned event-level eligibility, proximity, ranking and match-alert controls. **Owner:** Event admin/NLP **Classification:** Internal configuration **Retention:** Event life + audit period

Column	SQL type	Null	Key	Default	Description
event_matching_policy_id	bigint	No	PK	IDENTITY	Immutable policy version identifier.
event_id	varchar(64)	No	FK		FK Event; policy applies only to this event/context.
policy_version	smallint	No		1	Monotonic version within the event.
status	varchar(16)	No		'DRAFT'	DRAFT, ACTIVE or RETIRED.
registration_required	bit	No		1	Member must hold an eligible EventRegistration.
check_in_required	bit	No		0	Member must be checked in before event matching.
live_mode_required	bit	No		1	Active consent-backed LiveModeSession required.
proximity_mode	varchar(24)	No		'VENUE'	NONE, VENUE or COARSE_CELL.
max_presence_age_minutes	smallint	Yes		15	Maximum age of coarse presence when proximity is used.

Column	SQL type	Null	Key	Default	Description
match_threshold_override	decimal(6,5)	Yes			Optional override of active NLP ranking threshold.
alert_confidence_threshold	decimal(6,5)	No		0.70	Minimum final score before a proactive match alert is eligible.
max_match_alerts_per_hour	smallint	No		2	Event-specific hourly cap per member.
max_match_alerts_per_event	smallint	No		10	Event-lifetime cap per member.
minimum_alert_interval_minutes	smallint	No		30	Minimum spacing between event match alerts.
effective_from	datetimeoffset(7)	No			Start of policy applicability.
effective_to	datetimeoffset(7)	Yes			Optional retirement time.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: UQ(event_id,policy_version); filtered UQ one ACTIVE policy per event; IX(status,effective_from,effective_to)

Rules: CHECK thresholds 0..1, caps >= 0 and effective_to > effective_from. COARSE_CELL requires live_mode_required and a fresh EventPresence. Activation and retirement are audited; an event alert must satisfy this policy and notification policy.

event.EventRegistration

Purpose: Member eligibility/check-in relationship to an event. **Owner:** Event/Core API **Classification:** Personal **Retention:** Event + approved participation history

Column	SQL type	Null	Key	Default	Description
event_registration_id	bigint	No	PK	IDENTITY	Internal key.
event_id	varchar(64)	No	FK		FK Event.
member_id	varchar(64)	No	FK		FK Member.
status	varchar(24)	No		'REGISTERED'	INVITED, REGISTERED, CHECKED_IN, CANCELLED.
registered_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Registration time.
checked_in_at	datetimeoffset(7)	Yes			Event check-in time.
source	varchar(24)	No		'APP'	APP, ADMIN, IMPORT.

Column	SQL type	Null	Key	Default	Description
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: UQ(event_id,member_id); IX(member_id,status); IX(event_id,status)

Rules: Check-in is not consent for Live Mode; consent is evaluated separately.

event.LiveModeSession

Purpose: Bounded, revocable member discovery session. **Owner:** Event/Core API **Classification:** Sensitive location context **Retention:** Short: recommended 24-72 hours after event

Column	SQL type	Null	Key	Default	Description
live_session_id	varchar(64)	No	PK		Session identifier.
event_id	varchar(64)	No	FK		FK Event.
member_id	varchar(64)	No	FK		FK Member.
consent_record_id	bigint	No	FK		Valid Live Mode consent evidence.
status	varchar(16)	No		'ACTIVE'	ACTIVE, DISABLED, EXPIRED.
activated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Activation time.
active_until	datetimeoffset(7)	No			Hard expiry bounded by event.
disabled_at	datetimeoffset(7)	Yes			Immediate revocation time.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: IX(event_id,status,active_until); IX(member_id,status)

Rules: One active session per member/event via filtered unique index.

event.EventPresence

Purpose: Coarse, short-lived evidence used for nearby matching. **Owner:** Event proximity service **Classification:** Highly sensitive location **Retention:** Hours; maximum per approved policy

Column	SQL type	Null	Key	Default	Description
presence_id	bigint	No	PK	IDENTITY	Internal key.
live_session_id	varchar(64)	No	FK		FK LiveModeSession.
coarse_cell	varchar(32)	No			Approved coarse geohash/H3 cell.
observed_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Observation time.
expires_at	datetimeoffset(7)	No			Automatic purge time.
source	varchar(24)	No			CHECK_IN, FOREGROUND_GEO or VENUE_ZONE.

Indexes: IX(live_session_id,observed_at DESC); IX(coarse_cell,expires_at)

Rules: Never expose another member's coarse cell or raw observation time.

13. SOCIAL schema

social.ConnectionRequest

Purpose: Consent gate before creating a connection/chat. **Owner:** Social API **Classification:** Confidential **Retention:** 2 years or approved policy

Column	SQL type	Null	Key	Default	Description
connection_request_id	varchar(64)	No	PK		Request identifier.
sender_member_id	varchar(64)	No	FK		Requester.
recipient_member_id	varchar(64)	No	FK		Recipient.
context_id	varchar(64)	Yes			Event or general matching context.
match_result_id	bigint	Yes	FK		Recommendation that led to request.
note	nvarchar(500)	Yes			Optional introduction note; moderate as content.
status	varchar(24)	No		'PENDING'	PENDING, ACCEPTED, DECLINED, WITHDRAWN, EXPIRED.
expires_at	datetimeoffset(7)	No			Policy-calculated expiry for a pending request.
responded_at	datetimeoffset(7)	Yes			Terminal response time.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.

Column	SQL type	Null	Key	Default	Description
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: IX(recipient_member_id,status,expires_at,created_at DESC); IX(sender_member_id,status); filtered UQ open pair
Rules: CHECK sender <> recipient and expires_at > created_at. Block or existing connection prevents creation. An expiry worker changes overdue PENDING requests to EXPIRED idempotently.

social.Connection

Purpose: Mutually accepted member relationship. **Owner:** Social API **Classification:** Confidential **Retention:** Until disconnect + policy period

Column	SQL type	Null	Key	Default	Description
connection_id	varchar(64)	No	PK		Connection identifier.
member_low_id	varchar(64)	No	FK		Lexicographically lower member ID.
member_high_id	varchar(64)	No	FK		Lexicographically higher member ID.
accepted_request_id	varchar(64)	No	FK/UQ		Accepted ConnectionRequest.
status	varchar(20)	No		'ACTIVE'	ACTIVE or DISCONNECTED.
connected_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Acceptance time.
disconnected_at	datetimeoffset(7)	Yes			Relationship end.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: UQ(member_low_id,member_high_id); IX(member_low_id,status); IX(member_high_id,status)
Rules: Canonical pair columns prevent duplicate reverse relationships.

social.MemberBlock

Purpose: Directional block that suppresses discovery and communication both ways. **Owner:** Safety/Social API **Classification:** Highly restricted **Retention:** Until member removes + safety policy

Column	SQL type	Null	Key	Default	Description
block_id	bigint	No	PK	IDENTITY	Internal key.
blocker_member_id	varchar(64)	No	FK		Member initiating block.

Column	SQL type	Null	Key	Default	Description
blocked_member_id	varchar(64)	No	FK		Blocked member.
reason_code	varchar(64)	Yes			Private controlled reason.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Effective immediately.
removed_at	datetimeoffset(7)	Yes			Optional unblock.

Indexes: UQ active(blocker_member_id,blocked_member_id); IX(blocked_member_id,removed_at)

Rules: Never reveal who blocked whom through product APIs.

social.MemberReport

Purpose: Member-submitted safety/report workflow trigger. **Owner:** Safety/Moderation **Classification:** Highly restricted **Retention:** Per safety/legal policy

Column	SQL type	Null	Key	Default	Description
report_id	varchar(64)	No	PK		Report identifier.
reporter_member_id	varchar(64)	No	FK		Reporter.
reported_member_id	varchar(64)	No	FK		Subject.
resource_type	varchar(32)	Yes			PROFILE, REQUEST, MESSAGE, ATTACHMENT.
resource_id	varchar(64)	Yes			Reported object.
category	varchar(64)	No			Controlled report category.
description	nvarchar(2000)	Yes			Reporter description.
status	varchar(24)	No		'OPEN'	OPEN, TRIAGED, RESOLVED, DISMISSED.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: IX(status,created_at); IX(reported_member_id,created_at DESC)

Rules: Creates/links a ModerationCase; access limited to safety roles.

14. CHAT schema

chat.Conversation

Purpose: Accepted-connection one-to-one chat container. **Owner:** Chat service **Classification:** Confidential **Retention:** Connection life + message policy

Column	SQL type	Null	Key	Default	Description
conversation_id	varchar(64)	No	PK		Conversation identifier.
connection_id	varchar(64)	No	FK/UQ		One conversation per accepted connection.
status	varchar(20)	No		'ACTIVE'	ACTIVE, CLOSED, RESTRICTED.
last_message_at	datetimeoffset(7)	Yes			Conversation list ordering.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: UQ(connection_id); IX(status,last_message_at DESC)

Rules: Creation only in the same transaction/outbox flow as connection acceptance.

chat.ConversationParticipant

Purpose: Per-member chat state and authorization projection. **Owner:** Chat service **Classification:** Confidential **Retention:** Conversation life

Column	SQL type	Null	Key	Default	Description
conversation_id	varchar(64)	No	PK/FK		FK Conversation.
member_id	varchar(64)	No	PK/FK		Participant.
joined_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Participation start.
last_read_message_id	varchar(64)	Yes	FK		Read cursor.
muted_until	datetimeoffset(7)	Yes			Notification suppression.
left_at	datetimeoffset(7)	Yes			Closure/disconnect projection.

Indexes: PK(conversation_id,member_id); IX(member_id,left_at)

Rules: Exactly two participants and both must match active connection pair.

chat.Message

Purpose: Durable one-to-one message. **Owner:** Chat service **Classification:** Highly confidential **Retention:** Approved chat retention/deletion policy

Column	SQL type	Null	Key	Default	Description
message_id	varchar(64)	No	PK		Client-generated ID supports idempotency.
conversation_id	varchar(64)	No	FK		FK Conversation.
sender_member_id	varchar(64)	No	FK		Authorized participant.
message_type	varchar(20)	No		'TEXT'	TEXT, FILE, SYSTEM.
body	nvarchar(max)	Yes			Text content; sanitize for display.
client_sent_at	datetimeoffset(7)	Yes			Client timestamp for UX only.
server_sequence	bigint	No	UQ	SEQUENCE	Monotonic per conversation or global stream.
moderation_status	varchar(24)	No		'PENDING_OR_CLEAR'	Safety state.
deleted_at	datetimeoffset(7)	Yes			Logical removal time.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Authoritative send time.
row_version	rowversion	No	Concurrency	Generated	Concurrency token.

Indexes: UQ(conversation_id,message_id); IX(conversation_id,server_sequence); IX(sender_member_id,created_at)

Rules: Server verifies active connection and blocks on every send.

chat.Attachment

Purpose: Blob metadata and malware-scan lifecycle. **Owner:** Chat/File service **Classification:** Inherits file content **Retention:** Message/file policy

Column	SQL type	Null	Key	Default	Description
attachment_id	varchar(64)	No	PK		Attachment identifier.
message_id	varchar(64)	Yes	FK		Message after finalize; null during upload.
owner_member_id	varchar(64)	No	FK		Uploader.
blob_path	nvarchar(1024)	No	UQ		Private container path; never store SAS URL.
file_name	nvarchar(255)	No			Sanitized display name.

Column	SQL type	Null	Key	Default	Description
media_type	varchar(128)	No			Verified MIME type.
size_bytes	bigint	No			Validated size.
sha256	char(64)	No			Integrity/deduplication hash.
scan_status	varchar(24)	No		'PENDING'	PENDING, CLEAN, REJECTED, ERROR.
expires_at	datetimeoffset(7)	Yes			Orphan/temp purge time.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: UQ(blob_path); IX(owner_member_id,created_at); IX(scan_status,created_at)

Rules: Download authorization resolves current conversation membership and emits short-lived signed URL.

chat.MessageReceipt

Purpose: Message delivered/read evidence by participant. **Owner:** Chat service **Classification:** Confidential **Retention:** Message life

Column	SQL type	Null	Key	Default	Description
message_id	varchar(64)	No	PK/FK		FK Message.
member_id	varchar(64)	No	PK/FK		Recipient participant.
delivered_at	datetimeoffset(7)	Yes			First delivery.
read_at	datetimeoffset(7)	Yes			First read.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Last receipt update.

Indexes: PK(message_id,member_id); IX(member_id,read_at)

Rules: Updates are monotonic; timestamps never move backwards.

15. NOTIFICATION schema

notification.NotificationPolicy

Purpose: Versioned channel, frequency, quiet-hours, retry and expiry controls by notification purpose. **Owner:** Notification admin **Classification:** Internal configuration **Retention:** Indefinite/versioned

Column	SQL type	Null	Key	Default	Description
notification_policy_id	bigint	No	PK	IDENTITY	Immutable policy version identifier.
community_id	varchar(64)	Yes	FK		Optional community scope; null is platform default.
purpose_code	varchar(64)	No			MATCH, REQUEST, CHAT, EVENT, SAFETY or ACCOUNT.
channel	varchar(16)	No			PUSH, EMAIL or IN_APP.
policy_version	smallint	No		1	Monotonic version for scope/purpose/channel.
status	varchar(16)	No		'DRAFT'	DRAFT, ACTIVE or RETIRED.
member_opt_out_allowed	bit	No		1	Whether member preference may disable this notification.
quiet_hours_behavior	varchar(16)	No		'DEFER'	DEFER, SUPPRESS or BYPASS.
dedupe_window_seconds	int	No		300	Duplicate suppression window for the same dedupe key.
max_per_hour	smallint	No		6	Maximum queued/sent notifications per member and policy each hour.
max_per_day	smallint	No		30	Maximum queued/sent notifications per member and policy each day.
max_attempts	smallint	No		3	Bounded provider delivery attempts.
retry_schedule_seconds	varchar(128)	No		'60,300,1800'	Validated comma-separated retry delays.
ttl_minutes	int	No		1440	Notification expiry after initial eligibility.
effective_from	datetimeoffset(7)	No			Start of policy applicability.
effective_to	datetimeoffset(7)	Yes			Optional retirement time.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: UQ(community_id,purpose_code,channel,policy_version); filtered UQ one ACTIVE policy per scope/purpose/channel; IX(status,effective_from,effective_to)

Rules: CHECK limits and TTL > 0 and effective_to > effective_from. BYPASS and non-optional delivery are restricted to approved SAFETY/ACCOUNT purposes. Policies are immutable after activation; changes create a new version and AuditEvent.

notification.NotificationPreference

Purpose: Member channel, quiet hours and purpose preferences. **Owner:** Notification service **Classification:** Personal **Retention:** Account life

Column	SQL type	Null	Key	Default	Description
member_id	varchar(64)	No	PK/FK		FK Member.
purpose_code	varchar(64)	No	PK		MATCH, REQUEST, CHAT, EVENT, SAFETY or ACCOUNT.
push_enabled	bit	No		1	Push channel preference.
email_enabled	bit	No		0	Email channel preference.
quiet_start_local	time	Yes			Optional local quiet start.
quiet_end_local	time	Yes			Optional local quiet end.
timezone_id	varchar(64)	Yes			Required when quiet hours set.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: PK(member_id,purpose_code)

Rules: Safety/account notifications may be mandatory and are identified separately.

notification.PushToken

Purpose: Provider push token mapped to a registered device. **Owner:** Notification service **Classification:** Secret-like personal **Retention:** Until rotation/revocation + short grace

Column	SQL type	Null	Key	Default	Description
push_token_id	bigint	No	PK	IDENTITY	Internal key.
device_id	varchar(64)	No	FK		FK MemberDevice.
provider	varchar(24)	No			APNS or FCM.
token_ciphertext	varbinary(max)	No	UQ		Encrypted/tokenized push token.
status	varchar(16)	No		'ACTIVE'	ACTIVE or INVALID.

Column	SQL type	Null	Key	Default	Description
last_success_at	datetimeoffset(7)	Yes			Last accepted delivery.
invalidated_at	datetimeoffset(7)	Yes			Provider rejection/revocation.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: UQ token fingerprint; IX(device_id,status)

Rules: Restrict read access to notification service identity.

notification.Notification

Purpose: Policy-resolved, rate-limited delivery intent and aggregate outcome. **Owner:** Notification service **Classification:** Confidential **Retention:** 90 days default; confirm policy

Column	SQL type	Null	Key	Default	Description
notification_id	varchar(64)	No	PK		Delivery identifier.
member_id	varchar(64)	No	FK		Recipient.
notification_policy_id	bigint	No	FK		Immutable NotificationPolicy version applied.
event_matching_policy_id	bigint	Yes	FK		EventMatchingPolicy applied to an event match alert.
purpose_code	varchar(64)	No			Notification purpose.
channel	varchar(16)	No			Resolved PUSH, EMAIL or IN_APP channel.
resource_type	varchar(32)	Yes			MATCH, REQUEST, MESSAGE, EVENT.
resource_id	varchar(64)	Yes			Deep-link resource.
template_code	varchar(64)	No			Versioned content template.
dedupe_key	varchar(160)	No			Stable purpose/member/resource key used for duplicate suppression.
dedupe_bucket_start	datetimeoffset(7)	No			Start of the policy-derived deduplication window.
source_confidence	decimal(6,5)	Yes			Match confidence captured when threshold-based notification is used.
context_id	varchar(64)	Yes			Event ID or GENERAL context used for event-lifetime caps.

Column	SQL type	Null	Key	Default	Description
status	varchar(20)	No		'PENDING'	PENDING, SENT, DELIVERED, FAILED, SUPPRESSED.
scheduled_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Quiet-hours-aware schedule.
expires_at	datetimeoffset(7)	No			Policy-calculated time after which delivery is suppressed.
attempt_count	smallint	No		0	Aggregate count of append-only delivery attempts.
sent_at	datetimeoffset(7)	Yes			Provider acceptance time.
delivered_at	datetimeoffset(7)	Yes			Provider delivery acknowledgment when available.
suppression_reason	varchar(64)	Yes			OPT_OUT, QUIET_HOURS, RATE_LIMIT, DUPLICATE, EXPIRED, BELOW_THRESHOLD or POLICY.
failure_code	varchar(64)	Yes			Sanitized provider failure.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: UQ(member_id,channel,dedupe_key,dedupe_bucket_start); IX(status,scheduled_at,expires_at); IX(member_id,notification_policy_id,created_at); IX(member_id,event_matching_policy_id,context_id,created_at); IX(member_id,created_at DESC)

Rules: notification.TryEnqueue resolves active policy and member preference, checks event threshold, applies quiet hours, dedupe and hourly/daily caps atomically, and applies the stricter event-vs-channel limit before writing Notification and OutboxEvent. Payload contains minimum preview data; full content is fetched after authentication.

notification.NotificationDeliveryAttempt

Purpose: Append-only provider attempt and acknowledgment history. **Owner:** Notification worker **Classification:** Confidential operational metadata **Retention:** Notification life + 30 days

Column	SQL type	Null	Key	Default	Description
delivery_attempt_id	bigint	No	PK	IDENTITY	Attempt identifier.
notification_id	varchar(64)	No	FK		FK Notification.
attempt_number	smallint	No			One-based attempt number.
provider	varchar(32)	No			APNS, FCM, EMAIL_PROVIDER or IN_APP.
push_token_id	bigint	Yes	FK		PushToken used; null for non-push channels.
provider_message_id	nvarchar(256)	Yes			Provider correlation ID; never a credential.

Column	SQL type	Null	Key	Default	Description
status	varchar(20)	No		'STARTED'	STARTED, ACCEPTED, DELIVERED, FAILED or EXPIRED.
attempted_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Attempt start time.
acknowledged_at	datetimeoffset(7)	Yes			Provider acknowledgment time.
next_attempt_at	datetimeoffset(7)	Yes			Next policy-derived retry time.
duration_ms	int	Yes			Provider request duration.
failure_code	varchar(64)	Yes			Sanitized failure category.

Indexes: UQ(notification_id,attempt_number); IX(status,next_attempt_at); IX(provider,provider_message_id)

Rules: Append only. Attempts stop when delivered, expired or max_attempts is reached. Invalid-token responses revoke PushToken. Provider payloads and secrets are not stored.

16. NLP schema

nlp.NlpIntent

Purpose: Current WANT/OFFER text and structured matching metadata. **Owner:** NLP service **Classification:** Personal/business text **Retention:** Active intent + approved history

Column	SQL type	Null	Key	Default	Description
intent_id	varchar(64)	No	PK		Matches existing API contract.
member_id	varchar(64)	No	FK		Authoritative FK iam.Member.
context_id	varchar(64)	No			Event ID or GENERAL context.
intent_type	varchar(16)	No			WANT or OFFER.
original_text	nvarchar(4000)	No			Member-authored display text.
normalized_text	nvarchar(4000)	No			PII-minimized normalization.
normalized_hash	char(64)	No			Content hash for idempotent embedding.
language_code	varchar(16)	No		'en'	Detected/approved language.
contains_pii	bit	No		0	Preprocessing signal.
status	varchar(32)	No		'PROCESSING'	PROCESSING, MATCH_READY, FAILED, INACTIVE.

Column	SQL type	Null	Key	Default	Description
category	nvarchar(128)	Yes			Structured category.
industry	nvarchar(128)	Yes			Structured industry.
geography	nvarchar(128)	Yes			Business geography, not live location.
expires_at	datetimeoffset(7)	No			Freshness/eligibility expiry.
preprocessing_version	varchar(128)	No			Normalizer version.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: IX(context_id,intent_type,status,expires_at); IX(member_id,status); IX(normalized_hash)

Rules: Normalization is synchronous for MVP: original_text, normalized_text, normalized_hash, language_code and preprocessing_version are committed together. Embedding and ranking remain asynchronous.

nlp.NlpEmbedding

Purpose: Versioned vector for a normalized intent. **Owner:** NLP service **Classification:** Derived sensitive **Retention:** Intent life + evaluation period

Column	SQL type	Null	Key	Default	Description
intent_id	varchar(64)	No	PK/FK		FK NlpIntent.
model_version	varchar(128)	No	PK/FK		FK NlpModelVersion.
dimensions	int	No			Vector dimensions.
normalized_hash	char(64)	No			Text version embedded.
embedding	varbinary(max) / vector	No			Existing varbinary baseline; native vector after target validation.
status	varchar(20)	No		'ACTIVE'	ACTIVE, SUPERSEDED, FAILED.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Creation time.

Indexes: PK(intent_id,model_version); UQ(intent_id,normalized_hash,model_version); vector/ANN index when supported

Rules: One embedding space/model for both WANT and OFFER.

nlp.NlpModelVersion

Purpose: Embedding provider/deployment and preprocessing compatibility. **Owner:** NLP evaluator **Classification:** Internal configuration **Retention:** Indefinite/versioned

Column	SQL type	Null	Key	Default	Description
model_version	varchar(128)	No	PK		Stable version.
provider	varchar(64)	No			Approved provider.
deployment_name	varchar(128)	No			Configuration reference, not secret.
dimensions	int	No			Expected vector length.
preprocessing_version	varchar(128)	No			Compatible preprocessing.
status	varchar(20)	No		'CANDIDATE'	CANDIDATE, ACTIVE, RETIRED, ROLLED_BACK.
activated_at	datetimeoffset(7)	Yes			Promotion time.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Registration time.

Indexes: Filtered UQ one ACTIVE model; IX(status,created_at)

Rules: Promotion requires evaluation evidence and rollback path.

nlp.NlpRankingConfig

Purpose: Versioned ranking weights, threshold and policy switches. **Owner:** NLP evaluator **Classification:** Internal configuration **Retention:** Indefinite/versioned

Column	SQL type	Null	Key	Default	Description
ranking_version	varchar(128)	No	PK		Stable configuration version.
semantic_weight	decimal(6,5)	No		0.40	Semantic/reciprocal weight.
category_weight	decimal(6,5)	No		0.25	Category compatibility.
industry_weight	decimal(6,5)	No		0.15	Industry compatibility.
geography_weight	decimal(6,5)	No		0.10	Business geography fit.
freshness_weight	decimal(6,5)	No		0.10	Intent freshness.
event_weight	decimal(6,5)	No		0	Optional event context after evaluation.
threshold	decimal(6,5)	No		0.35	Display threshold.

Column	SQL type	Null	Key	Default	Description
active_from	datetimeoffset(7)	No			Effective time.
active_to	datetimeoffset(7)	Yes			Retirement time.
config_json	nvarchar(2000)	Yes			Bounded extra rule settings.

Indexes: IX(active_from,active_to); filtered UQ active interval

Rules: CHECK weights 0..1 and approved sum; immutable after activation.

nlp.NlpProcessingJob

Purpose: Intent embedding and re-embedding retry state. **Owner:** NLP worker **Classification:** Confidential **Retention:** 30-90 days after completion

Column	SQL type	Null	Key	Default	Description
job_id	varchar(64)	No	PK		Job identifier.
intent_id	varchar(64)	No	FK		Target intent.
job_type	varchar(24)	No		'EMBED'	EMBED or REEMBED. Normalization is synchronous for MVP.
status	varchar(20)	No		'PENDING'	PENDING, RUNNING, SUCCEEDED, FAILED, DEAD.
attempt_count	int	No		0	Bounded attempts.
available_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Retry schedule.
locked_until	datetimeoffset(7)	Yes			Worker lease.
error_code	varchar(64)	Yes			Sanitized failure code.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: UQ active(intent_id,job_type); IX(status,available_at)

Rules: Do not persist raw provider errors that may include content.

nlp.MatchRequest

Purpose: Idempotent match-search execution envelope. **Owner:** NLP service **Classification:** Confidential **Retention:** 90 days default; confirm

Column	SQL type	Null	Key	Default	Description
request_id	varchar(64)	No	PK		Client/API idempotency key.
request_hash	char(64)	No			Canonical hash of matching inputs; detects idempotency-key reuse with different inputs.
requester_id	varchar(64)	No	FK		Authenticated member.
intent_id	varchar(64)	No	FK		Requester intent.
context_id	varchar(64)	No			Matching context.
language_code	varchar(16)	No		'en'	Language snapshot used for this request.
requested_limit	smallint	No		7	Allowed 3-7.
request_options_json	nvarchar(2000)	Yes			Validated bounded options snapshot; no free-form secrets or raw provider payloads.
status	varchar(20)	No		'PROCESSING'	PROCESSING, COMPLETED, FAILED.
preprocessing_version	varchar(128)	Yes			Resolved synchronous normalization version.
model_version	varchar(128)	Yes	FK		Resolved model version.
ranking_version	varchar(128)	Yes	FK		Resolved ranking version.
ranking_threshold	decimal(6,5)	Yes			Immutable threshold snapshot applied to this execution.
candidate_count	int	Yes			Eligible bounded candidates examined.
completed_at	datetimeoffset(7)	Yes			Completion time.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: IX(requester_id,created_at DESC); IX(status,created_at); IX(request_hash)

Rules: Repeated request_id with the same request_hash returns prior result or current status; a different hash returns conflict. COMPLETED requires preprocessing_version, model_version, ranking_version, ranking_threshold and completed_at.

nlp.NlpMatchResult

Purpose: Ranked, explained recommendation returned for one request. **Owner:** NLP service **Classification:** Confidential/derived **Retention:** 90 days or product history policy

Column	SQL type	Null	Key	Default	Description
match_result_id	bigint	No	PK	IDENTITY	Internal key.
request_id	varchar(64)	No	FK		FK MatchRequest.
requester_id	varchar(64)	No	FK		Requester.
candidate_id	varchar(64)	No	FK		Candidate.
rank	smallint	No			1-based result rank.
semantic_score	decimal(8,7)	No			Semantic score.
reciprocal_score	decimal(8,7)	Yes			Bidirectional harmonic/approved score.
final_score	decimal(8,7)	No			Versioned final ranking score.
label	varchar(32)	No			STRONG_MATCH, PLAUSIBLE_MATCH, etc.
reason_codes	nvarchar(1000)	No			JSON array of approved codes.
reason_text	nvarchar(2000)	No			Deterministic explanation.
model_version	varchar(128)	No	FK		Embedding version.
preprocessing_version	varchar(128)	No			Normalization version.
ranking_version	varchar(128)	No	FK		Ranking configuration.
policy_status	varchar(24)	No		'ELIGIBLE'	ELIGIBLE or SUPPRESSED.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Creation time.

Indexes: UQ(request_id,candidate_id); UQ(request_id,rank); IX(candidate_id,created_at DESC)

Rules: A score never bypasses eligibility; suppressed results are not returned to members.

nlp.NlpFeedback

Purpose: Useful/not useful/inappropriate signal for recommendation quality. **Owner:** NLP service **Classification:** Confidential **Retention:** Evaluation policy; de-identify for long-term use

Column	SQL type	Null	Key	Default	Description
feedback_id	bigint	No	PK	IDENTITY	Internal key.
supersedes_feedback_id	bigint	Yes	FK/UQ		Optional self-FK to the earlier feedback corrected by this row.

Column	SQL type	Null	Key	Default	Description
match_result_id	bigint	No	FK		FK NlpMatchResult.
request_id	varchar(64)	No	FK		Denormalized request for compatibility.
requester_id	varchar(64)	No	FK		Authenticated feedback author.
candidate_id	varchar(64)	No	FK		Candidate.
label	varchar(64)	No			USEFUL, NOT_USEFUL, INAPPROPRIATE.
reason_code	varchar(64)	Yes			Controlled error category.
reason	nvarchar(1000)	Yes			Optional free text, PII checked.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Feedback time.

Indexes: UQ(match_result_id,requester_id) WHERE supersedes_feedback_id IS NULL; UQ(supersedes_feedback_id) WHERE supersedes_feedback_id IS NOT NULL; IX(label,created_at); IX(request_id)

Rules: Do not overwrite. A correction appends a row that self-references the immediately previous feedback; service validation prevents cycles and requires the same requester and match result.

nlp.MatchSuppression

Purpose: Administrative or safety suppression independent of model score. **Owner:** Moderation/NLP **Classification:** Restricted **Retention:** Until expiry + audit period

Column	SQL type	Null	Key	Default	Description
suppression_id	bigint	No	PK	IDENTITY	Internal key.
member_id	varchar(64)	Yes	FK		Affected member.
intent_id	varchar(64)	Yes	FK		Affected intent.
context_id	varchar(64)	Yes			Optional context.
reason_code	varchar(64)	No			Controlled reason.
starts_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Effective time.
ends_at	datetimeoffset(7)	Yes			Optional expiry.
created_by	varchar(64)	No	FK		Administrator/service actor.

Indexes: IX(member_id,context_id,ends_at); IX(intent_id,ends_at)

Rules: At least one target column must be non-null; changes are audited.

nlp.EvaluationDataset

Purpose: Versioned labelled-pair dataset metadata. **Owner:** NLP evaluator **Classification:** De-identified research **Retention:** Indefinite/versioned

Column	SQL type	Null	Key	Default	Description
dataset_id	varchar(64)	No	PK		Dataset version ID.
name	nvarchar(200)	No			Display name.
version	varchar(32)	No			Immutable version.
description	nvarchar(1000)	Yes			Sampling/labeling notes.
source_policy	nvarchar(500)	No			De-identification/provenance statement.
status	varchar(20)	No		'DRAFT'	DRAFT, APPROVED, RETIRED.
approved_by	varchar(64)	Yes	FK		Evaluator/admin.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: UQ(name,version); IX(status,created_at)

Rules: Never use raw production chat or exact location data.

nlp.EvaluationPair

Purpose: One labelled reciprocal matching example. **Owner:** NLP evaluator **Classification:** De-identified research **Retention:** Dataset life

Column	SQL type	Null	Key	Default	Description
evaluation_pair_id	bigint	No	PK	IDENTITY	Pair identifier.
dataset_id	varchar(64)	No	FK		FK EvaluationDataset.
requester_intent_text	nvarchar(4000)	No			De-identified requester sample.
candidate_intent_text	nvarchar(4000)	No			De-identified candidate sample.
structured_features_json	nvarchar(2000)	Yes			Category/industry/geography features.
gold_label	varchar(32)	No			STRONG, PLAUSIBLE, WEAK, NONE, UNSAFE.

Column	SQL type	Null	Key	Default	Description
split	varchar(16)	No			TRAIN, VALIDATION or TEST.
organization_group	varchar(64)	Yes			Leakage-prevention grouping, pseudonymous.
label_reason	nvarchar(1000)	Yes			Evaluator rationale.

Indexes: IX(dataset_id,split); IX(gold_label)

Rules: Pairs from the same member/organization remain in one split.

nlp.EvaluationRun

Purpose: Reproducible quality evaluation output. **Owner:** NLP evaluator **Classification:** Internal metrics **Retention:** Indefinite/versioned

Column	SQL type	Null	Key	Default	Description
evaluation_run_id	varchar(64)	No	PK		Run identifier.
dataset_id	varchar(64)	No	FK		Dataset version.
model_version	varchar(128)	No	FK		Model under test.
ranking_version	varchar(128)	No	FK		Ranking config.
status	varchar(20)	No		'RUNNING'	RUNNING, PASSED, FAILED, ERROR.
metrics_json	nvarchar(max)	Yes			Precision@5, Recall@20, reciprocal precision, coverage.
error_report_blob_path	nvarchar(1024)	Yes			Private report artifact.
started_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Run start.
completed_at	datetimeoffset(7)	Yes			Run completion.

Indexes: IX(dataset_id,started_at DESC); IX(model_version,ranking_version)

Rules: Activation gate references an approved run.

17. MODERATION schema

moderation.ModerationCase

Purpose: Unified case for member reports, content flags and account review. **Owner:** Moderation service **Classification:** Highly restricted **Retention:** Per safety/legal policy

Column	SQL type	Null	Key	Default	Description
moderation_case_id	varchar(64)	No	PK		Case identifier.
source_type	varchar(32)	No			MEMBER_REPORT, AUTO_SCAN, ADMIN_REVIEW.
source_id	varchar(64)	Yes			Source record identifier.
subject_member_id	varchar(64)	Yes	FK		Member under review.
resource_type	varchar(32)	Yes			PROFILE, MESSAGE, ATTACHMENT, INTENT.
resource_id	varchar(64)	Yes			Reviewed resource.
priority	varchar(16)	No		'NORMAL'	LOW, NORMAL, HIGH, URGENT.
status	varchar(24)	No		'OPEN'	OPEN, TRIAGED, ACTIONED, CLOSED.
assigned_to	varchar(64)	Yes	FK		Moderator.
closed_at	datetimeoffset(7)	Yes			Closure time.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: IX(status,priority,created_at); IX(subject_member_id,created_at DESC)

Rules: Access is separate from normal administration; views redact reporter identity where possible.

moderation.ModerationAction

Purpose: Append-only action taken within a moderation case. **Owner:** Moderation service **Classification:** Highly restricted **Retention:** Case retention

Column	SQL type	Null	Key	Default	Description
moderation_action_id	bigint	No	PK	IDENTITY	Action identifier.
moderation_case_id	varchar(64)	No	FK		FK ModerationCase.
actor_member_id	varchar(64)	No	FK		Moderator/admin.
action_type	varchar(64)	No			WARN, HIDE, SUSPEND, REJECT_FILE, CLOSE, etc.
reason_code	varchar(64)	No			Controlled reason.

Column	SQL type	Null	Key	Default	Description
notes	nvarchar(2000)	Yes			Restricted case notes.
effective_until	datetimeoffset(7)	Yes			Temporary action expiry.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Action time.

Indexes: IX(moderation_case_id,created_at); IX(actor_member_id,created_at)

Rules: Append-only; reversal is a new action.

moderation.ContentRule

Purpose: Versioned blocked-content and moderation configuration. **Owner:** Safety admin **Classification:** Internal configuration **Retention:** Indefinite/versioned

Column	SQL type	Null	Key	Default	Description
content_rule_id	varchar(64)	No	PK		Rule ID.
rule_type	varchar(32)	No			FILE_TYPE, TERM, RATE, POLICY.
version	varchar(32)	No			Rule version.
config_json	nvarchar(max)	No			Validated rule configuration.
status	varchar(16)	No		'DRAFT'	DRAFT, ACTIVE, RETIRED.
active_from	datetimeoffset(7)	Yes			Activation time.
created_by	varchar(64)	No	FK		Administrator.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: UQ(rule_type,version); IX(rule_type,status)

Rules: Configuration changes require review and audit.

moderation.ContentScan

Purpose: Malware/content scan result for files or text resources. **Owner:** File/Safety service **Classification:** Restricted **Retention:** Resource life + security period

Column	SQL type	Null	Key	Default	Description
content_scan_id	bigint	No	PK	IDENTITY	Scan record.
resource_type	varchar(32)	No			ATTACHMENT, PROFILE, INTENT, MESSAGE.
resource_id	varchar(64)	No			Target object.
scanner	varchar(64)	No			Scanner/provider ID.
scanner_version	varchar(64)	Yes			Engine/signature version.
result	varchar(24)	No			CLEAN, FLAGGED, MALICIOUS, ERROR.
reason_codes	nvarchar(1000)	Yes			Sanitized JSON reason codes.
scanned_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Scan time.

Indexes: IX(resource_type,resource_id,scanned_at DESC); IX(result,scanned_at)

Rules: Raw malicious content is not copied into this table.

18. OPS schema

ops.OutboxEvent

Purpose: Transactional event publication without dual-write loss. **Owner:** Owning service **Classification:** Internal/confidential **Retention:** 30-90 days after publish

Column	SQL type	Null	Key	Default	Description
outbox_event_id	varchar(64)	No	PK		Event ID/idempotency token.
aggregate_type	varchar(64)	No			MEMBER, CONNECTION, MESSAGE, INTENT.
aggregate_id	varchar(64)	No			Source aggregate.
event_type	varchar(128)	No			Versioned event name.
payload_json	nvarchar(max)	No			Minimal event payload; avoid raw content.
occurred_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Business event time.
published_at	datetimeoffset(7)	Yes			Successful publish time.
attempt_count	int	No		0	Publish attempts.

Column	SQL type	Null	Key	Default	Description
next_attempt_at	datetimeoffset(7)	Yes			Retry schedule.

Indexes: IX(published_at,next_attempt_at); IX(aggregate_type,aggregate_id,occurred_at)

Rules: Written in same transaction as business change.

ops.IdempotencyRecord

Purpose: Safe replay of mobile/API commands. **Owner:** API platform **Classification:** Confidential metadata **Retention:** 24 hours-30 days by operation

Column	SQL type	Null	Key	Default	Description
scope	varchar(64)	No	PK		API/operation scope.
idempotency_key	varchar(128)	No	PK		Client key.
actor_id	varchar(64)	No			Authenticated member/service.
request_hash	char(64)	No			Detect key reuse with different body.
status_code	smallint	Yes			Cached response status.
response_ref	nvarchar(1000)	Yes			Bounded response or resource reference.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	First request.
expires_at	datetimeoffset(7)	No			Purge time.

Indexes: PK(scope,idempotency_key); IX(expires_at)

Rules: Same key + different request hash returns conflict.

ops.BackgroundJob

Purpose: Non-NLP background task lifecycle. **Owner:** Platform workers **Classification:** Internal/confidential **Retention:** 30-90 days after completion

Column	SQL type	Null	Key	Default	Description
background_job_id	varchar(64)	No	PK		Job ID.
job_type	varchar(64)	No			SCAN_FILE, SEND_NOTIFICATION, PURGE_PRESENCE, etc.
resource_type	varchar(32)	Yes			Target type.
resource_id	varchar(64)	Yes			Target ID.

Column	SQL type	Null	Key	Default	Description
payload_json	nvarchar(2000)	Yes			Minimal validated payload.
status	varchar(20)	No		'PENDING'	PENDING, RUNNING, SUCCEEDED, FAILED, DEAD.
attempt_count	int	No		0	Attempts.
available_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Schedule/retry time.
error_code	varchar(64)	Yes			Sanitized failure.
created_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC creation time.
updated_at	datetimeoffset(7)	No		SYSUTCDATETIME()	UTC last material update time.
row_version	rowversion	No	Concurrency	Generated	Optimistic concurrency token; never client supplied.

Indexes: IX(status,available_at); IX(resource_type,resource_id)

Rules: Use Service Bus as transport when enabled; SQL row is durable business/job state, not a competing queue.

ops.AuditEvent

Purpose: Append-only evidence of sensitive/security/admin actions. **Owner:** Platform/security **Classification:** Highly restricted **Retention:** Per security/legal policy

Column	SQL type	Null	Key	Default	Description
audit_event_id	bigint	No	PK	IDENTITY	Monotonic audit key.
actor_type	varchar(24)	No			MEMBER, ADMIN, SERVICE, SYSTEM.
actor_id	varchar(64)	Yes			Pseudonymous/service identity.
action	varchar(128)	No			Versioned action code.
resource_type	varchar(64)	Yes			Affected type.
resource_id	varchar(64)	Yes			Affected ID.
outcome	varchar(16)	No			SUCCESS or DENIED/FAILED.
correlation_id	varchar(64)	No			End-to-end trace ID.
metadata_json	nvarchar(2000)	Yes			Allowlisted, content-free metadata.
occurred_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Event time.

Indexes: IX(resource_type,resource_id,occurred_at); IX(actor_id,occurred_at); IX(correlation_id)

Rules: Append-only permissions; no raw message, intent, token or location content.

19. ANALYTICS schema

analytics.ProductEvent

Purpose: Privacy-safe product funnel and match-quality analytics. **Owner:** Analytics pipeline **Classification:** Pseudonymous **Retention:** 13 months default; confirm policy

Column	SQL type	Null	Key	Default	Description
product_event_id	varchar(64)	No	PK		Event ID.
event_name	varchar(128)	No			PROFILE_COMPLETED, MATCH_VIEWED, REQUEST_SENT, etc.
member_pseudonym	char(64)	Yes			Environment-specific salted hash, not member_id.
community_id	varchar(64)	No			Aggregation boundary.
context_type	varchar(32)	Yes			EVENT or GENERAL.
context_id	varchar(64)	Yes			Allowed context reference.
properties_json	nvarchar(2000)	Yes			Allowlisted non-content dimensions.
occurred_at	datetimeoffset(7)	No			Client/server event time.
received_at	datetimeoffset(7)	No		SYSUTCDATETIME()	Server receipt time.

Indexes: IX(event_name,occurred_at); IX(community_id,occurred_at)

Rules: Prefer export to analytics service; keep raw text, contact, message and precise location out.

20. Indexing and performance strategy

- Index for actual API access paths: active profile discovery, context eligibility, pending requests, conversation sequence, active intents, match request/result lookup and due jobs.
- Use filtered indexes for active/pending rows. Avoid indexing free-text, JSON and high-churn timestamps without a measured query need.
- Candidate retrieval first applies eligibility filters, then vector similarity, and returns only 50-200 candidates for .NET ranking. Never scan the full member population in application memory.
- Use Query Store, execution-plan review and load tests before adding caches, partitioning or denormalized read models.
- Partition EventPresence, AuditEvent, ProductEvent and Message only when measured volume/maintenance windows justify it; begin with purge-friendly date indexes.

21. Mobile synchronization design

Concern	Design rule
Local data	Encrypted SQLite contains recent profile/match/conversation projections, sync cursor and pending commands only.
Writes	Client generates idempotency key/resource ID; server remains authoritative and returns version/ETag.
Conflicts	Optimistic concurrency. Server wins for match/consent/moderation; explicit merge only for approved editable profile fields.
Deletion	Server emits tombstone/removal event; app purges content after acknowledgement and on logout/device revocation.
Messages	Client message_id prevents duplicates; server_sequence establishes ordering.
Sensitive state	Tokens stay in OS keystore. Presence, moderation and administrative data are not cached beyond immediate UX need.

22. Security, privacy and retention

Control	Required baseline
Network	Private endpoint for Azure SQL/Storage where feasible; deny broad public access; isolate dev, test and production.
Identity	Managed identities for services; strong admin authentication; separate DB users/roles per service and environment.
Data	TDE at rest, TLS in transit, Key Vault secrets, encrypted/tokenized identity and push-token fields where required.
Authorization	Object-level checks on every profile, request, conversation, attachment and admin operation; never rely on client filtering.

Control	Required baseline
Location	Coarse EventPresence only while Live Mode is active; short retention and immediate revocation.
Testing	Synthetic or formally masked data only; no uncontrolled production copies in QA.
Audit	Append-only evidence for privileged access, consent, exports, deletion, moderation and model/ranking activation.

23. Existing NLP schema migration

Current object	Action	Target state
nlp.NlpIntent	Extend	Add normalized_hash, language_code, contains_pii and preprocessing_version; normalize synchronously before committing the intent; FK to iam.Member.
nlp.NlpEmbedding	Extend/validate	Keep composite PK; validate native Azure SQL vector type/index before migrating from varbinary.
nlp.NlpMatchResult	Normalize	Add MatchRequest FK, rank and component scores; retain request/candidate uniqueness.
nlp.MatchRequest	Add/extend	Persist request hash, language/options snapshot, preprocessing/model/ranking versions and applied threshold for reproducibility.
nlp.NlpFeedback	Extend	Reference match_result_id, controlled reason_code and self-referencing supersedes_feedback_id; retain existing request/member/candidate fields for compatibility.
nlp.MemberContextEligibility	Replace in integrated environment	Create nlp.vw_MemberContextEligibility over authoritative product tables.
nlp.MemberRelationship	Replace in integrated environment	Create nlp.vw_MemberRelationship over Connection and MemberBlock.
Stored procedures	Revise	Use views, enforce bounded results and return only fields required by CandidateRepository.

24. Delivery and QA acceptance gates

Gate	Evidence required by QA handover
Schema	All migrations build an empty Azure SQL database and upgrade the previous baseline repeatably.
Seed/reference data	Roles, sectors, consent policies, ranking config, event matching policies, notification policies and test events are versioned and deterministic.
Authorization	Blocked/unconnected pairs cannot match/chat/download; admin endpoints reject normal members.
Consent/Live Mode	Grant, deny, revoke, expire and purge paths pass; no exact location is returned.
Event alerts	Registration/check-in/Live Mode/proximity combinations, confidence thresholds, minimum intervals and per-hour/per-event caps pass, including immediate Live Mode revocation.

Gate	Evidence required by QA handover
Notifications	Opt-out, mandatory-purpose restrictions, quiet-hour defer/suppress, dedupe, hourly/daily limits, TTL, retries, invalid-token handling and provider acknowledgments pass under concurrent enqueue tests.
NLP	Intent processing, stored embedding reuse, eligibility views, reciprocal ranking, reasons and feedback are traceable.
Connections/chat	Acceptance creates one canonical connection/conversation; retry does not duplicate; files require clean scan.
Operations	Outbox/retry/dead-letter, monitoring, backup status, migration log and runbook are available.
Data quality	FK, unique, check, concurrency and retention tests pass; no critical/blocker defects remain.

25. Decisions still requiring product approval

1. Confirm community/tenant scope for Release 1; this draft supports one OLGA community without full multi-tenant isolation.
2. Approve retention periods for identity, chat, attachments, NLP inputs/results, feedback, audit and privacy cases.
3. Confirm whether event registration is mandatory before Live Mode and whether foreground coarse geolocation is required in QA.
4. Approve Release 1 event-alert defaults: 0.70 confidence threshold, two match alerts per hour, ten per event and a 30-minute minimum interval.
5. Approve notification purpose/channel defaults, hourly/daily caps, TTLs and which SAFETY/ACCOUNT messages may bypass opt-out or quiet hours.
6. Approve profile field visibility matrix before and after connection acceptance.
7. Confirm supported identity provider and whether email, mobile, or both are required for October 15.
8. Validate Azure SQL native vector support/compatibility in the target subscription before changing the working varbinary implementation.

Architect's immediate action Review Sections 6, 7 and 23 with backend, mobile, NLP, DevOps, security and QA leads. Freeze Must-have tables and transaction rules, then generate versioned SQL migrations and API contracts from this design.